

Jiwon Ryu

jiyajiwon.github.io [jiyajiwon](#)

Research Profile

- Incoming M.S. student in Computer Science at NYU Courant (Fall 2026).
- Interested in robot learning, manipulation, imitation learning, calibration/perception, real-world deployment, and human-in-the-loop correction.

Education

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| M.S. | New York University, Courant Institute of Mathematical Sciences , Computer Science <ul style="list-style-type: none"> • Incoming student, Fall 2026. | New York, NY |
| Under-graduate Study | Sejong University , Computer Science <ul style="list-style-type: none"> • Robotics / computer vision research internship experience. | Seoul, South Korea |

Experience

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| Robotics and Computer Vision Lab, Sejong University | Research Intern <ul style="list-style-type: none"> • Worked across real robot manipulation pipelines, including teleoperation, dataset collection, training, deployment, and evaluation. • Adapted calibration and robot-learning tooling to Piper and SO-101 hardware. • Debugged camera calibration, policy deployment behavior, action latency, and hardware/software integration issues. • Analyzed policy failures under camera changes, wrist-camera occlusion, OOD object placement, and grasp instability. | Seoul, South Korea
Jan 2026 – Apr 2026
4 months |
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Selected Projects

Real Robot Imitation Learning Pipeline

- Collected SO-101 demonstrations, fine-tuned a smolVLA model, and evaluated deployment under camera and object-placement changes.

Diffusion Policy Deployment and Failure Analysis

- Collected 100 pick-and-place episodes, deployed a trained checkpoint, and analyzed action-chunk latency and representation issues.

Calibration and Cross-View Alignment

- Adapted RAIDEN calibration tooling to Piper and validated wrist/scene camera transforms with Open3D.

Human-in-the-Loop Policy Correction

- Implemented joint-space delta intervention for corrective control during policy execution.

Technical Skills

Robot Learning: imitation learning, diffusion policy pipelines, smolVLA / VLA models, deployment evaluation

Robot Systems: Piper arm, SO-101 arm, leader-follower teleoperation, Jetson, CAN, RealSense

Calibration / Perception: hand-eye calibration, camera extrinsics, ChArUco, Open3D, multi-view alignment

Tools: Python, PyTorch, LeRobot, Hugging Face Hub, wandb, Linux